

1. Briefly describe the following terms connected with sky-wave propagation :

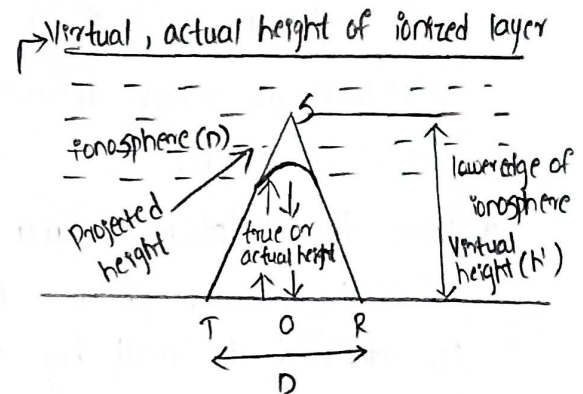
- a) Virtual height b) Critical frequency c) Maximum usable frequency d) Skip distance

a) Virtual height

EM waves are returned back to earth by successive refraction, that is they are down gradually rather sharply. Thus, this actual path of the wave ionized layer is given.

The height OS is called virtual height of the ionized layer as it is not the true height.

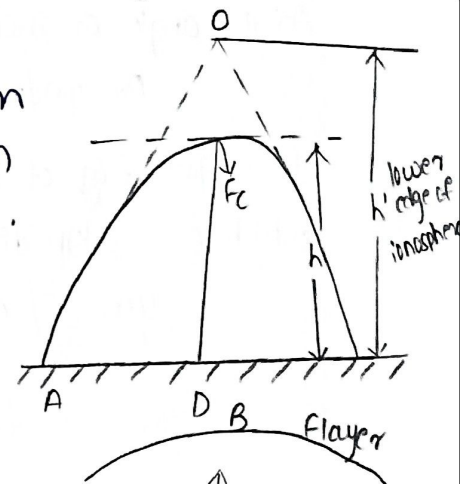
True height is OM. The wave being reflected rather than refracted, the path can be assumed to the straight line TS and RS.



b) Critical frequency

The highest frequency wave that will be reflected from given layer will be determined by maximum electron density N_{max} of that layer. Replace N by N_{max} and

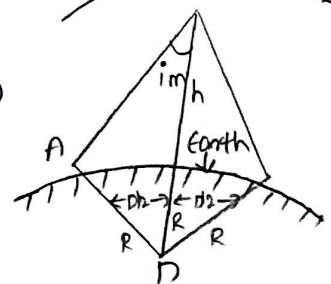
$$f_p = f_c \quad f_c = 9\sqrt{N_{max}}$$



c) Maximum Usable frequency

The maximum possible value of frequency for which reflection takes place for a given distance of propagation is called the Maximum usable frequency.

$$u = \frac{\sin i}{\sin 90^\circ} = \sqrt{1 - \frac{51.4m}{f^2 muf}}$$



d) Skip distance

The min distance from the transmitter at which sky wave of given frequency is returned to earth by ionosphere is called skip distance.

2. Write a short note on multi-hop propagation.

M curves is known as modified index curves. These curves show the variations of refractive index with height. In order to account for the curvature of earth, the actual index of refraction is modified to another refractive index.

Due to change in index of refraction the straight rays are converted to curved rays above flat earth. M curves are used to predict the type of transmission path for propagation of EM waves.

Characteristics of M-curves:

- i) standard propagation occurs, when modified index of refraction linearly varies with height, m-curve is a straight line having positive slope.
- ii) In order to achieve greater coverage the slope the m-curve increases near surface of earth.

3. Derive the relation between Maximum Usable Frequency (MUF) and skip distance. It is the minimum distance from transmitter at which skywave of given frequency is returned to earth by ionosphere

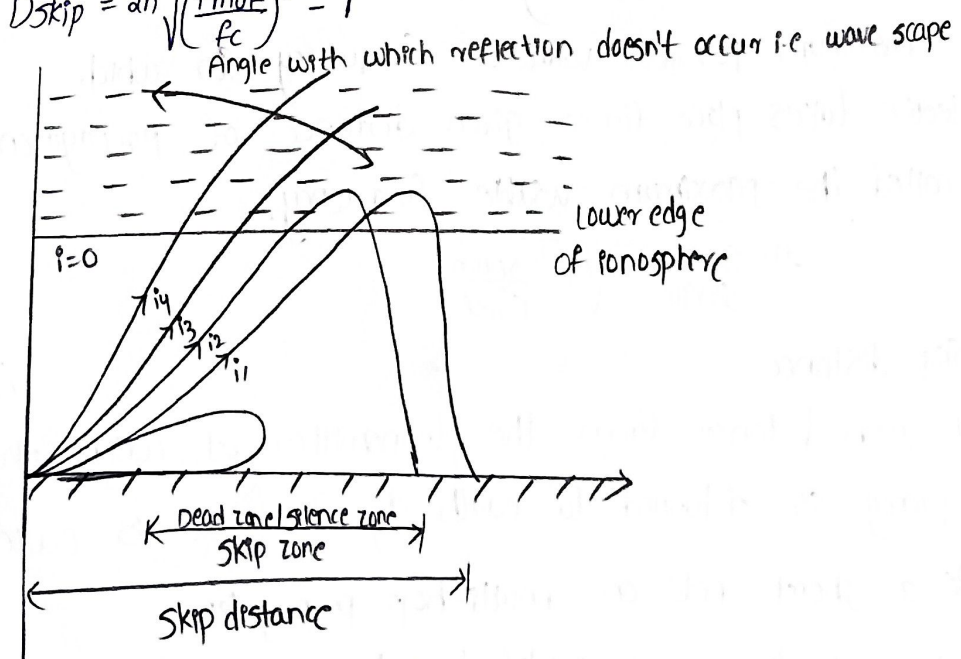
As the angle of incidence ' i ' is large. The mean n is slightly less than unity

$$n = \sin \phi = \sqrt{1 - \frac{8N}{f^2}}$$

For a given freq of propagation $f = f_{MUF}$ the skip distance can be calculated in which θ is skip distance

$$\frac{f_{MUF}}{f_c} = \sqrt{1 + \left(\frac{D}{2h}\right)^2} \Rightarrow \frac{f_{MUF}}{f_c} - 1 = \left(\frac{D_{skip}}{2h}\right)^2$$

$$D_{skip} = 2h \sqrt{\left(\frac{f_{MUF}}{f_c}\right)^2 - 1}$$



4. What is reciprocity of an antenna?

The reciprocity theorem states that the transmitting and receiving characteristics of an antenna are identical.

This means an antenna that performs well as a transmitting antenna will perform equally well as a receiving antenna under the same conditions.

Key points

- The radiation pattern of an antenna is the same for transmission and reception.
- Impedance, gain and directivity remain unchanged
- If antenna A transmits to antenna B, then B can transmit the same signal back to A with the same efficiency.

Ex:

If a dipole antenna can receive signals effectively from a direction, it can also transmit signals effectively in that direction.

5. What is near field and far field region? why is the condition $2D^2/\lambda$ chosen for far field region.

1. Near Field Region

The region close to the antenna where the EM field pattern is not fully formed

Characteristics:

- Reactive energy dominates
- Electric and magnetic fields are not perpendicular

Near field Region is also called Fresnel Region

2. Far Field Region

The region far from the antenna where the radiation pattern becomes stable.

Characteristics:

- Electric field and magnetic field are perpendicular.
- waves behave like plane waves.

This region is also called Fraunhofer region.

Condition $R \geq \frac{2D^2}{\lambda}$ is chosen:

Where D = max dimension of antenna

λ = wavelength

R = distance from antenna

The condition $R \geq \frac{2D^2}{\lambda}$ ensures that

- the wavefront becomes plane
- the radiation pattern becomes independent of distance.

6. With a neat sketch explain the procedure of radiation pattern measurement.

- It is measured at fix distance r and at fix freq. f

- Here angle θ varies from 0 to π

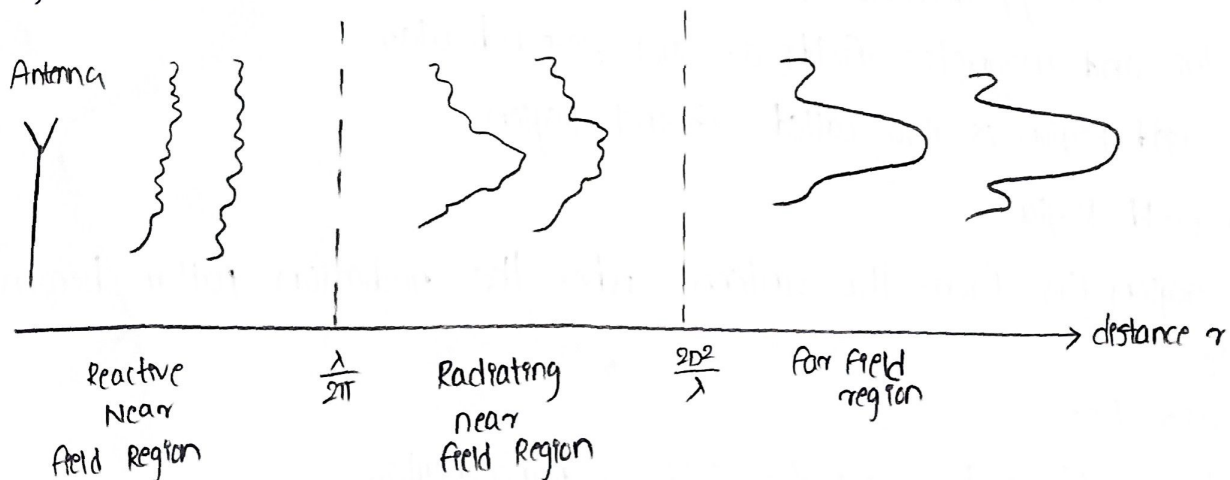
- Here angle ϕ varies from 0 to 2π

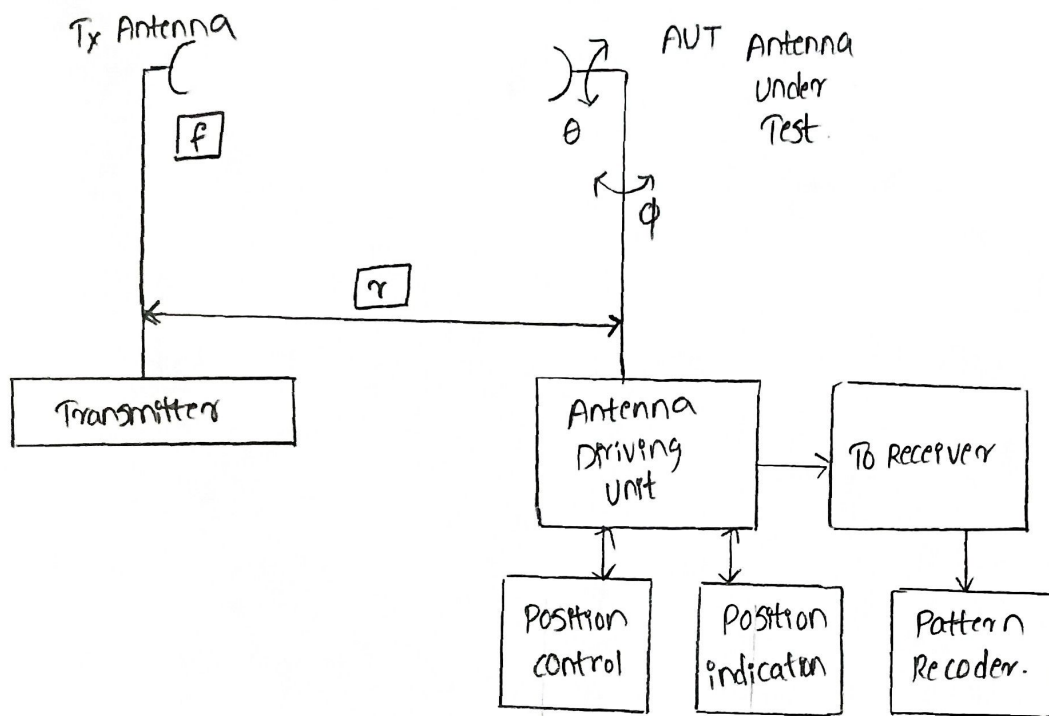
Definition:

- A plot of radiation characteristics of antenna as a function of θ and ϕ for constant distance r and frequency f is called as the Radiation pattern of the antenna.

Parameters:

FNBW	side lobes	Front to back ratio.
HFBW	Nulls	
Major lobes	Maxima	





7. Explain the procedure of antenna gain measurement.

Antenna gain is measured by comparing the test antenna with a standard antenna whose gain is already known.

Procedure:

1. place a standard transmitting antenna at a known distance.
2. Measure the received power using a reference antenna
3. Replace the reference antenna with the antenna under test (AUT)
4. measure the received power again.
5. Using the power ratio, calculate the gain of the test antenna.

Gain formula : $G_{test} = G_{ref} + 10 \log \left(\frac{P_{test}}{P_{ref}} \right)$

where

G_{test} = Gain of test antenna

G_{ref} = Gain of reference antenna

P_{test} = Power received by test antenna

P_{ref} = Power received by reference antenna.